

Kontakt

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Top-Kenntnisse

Digitale Transformation
Analytische Fähigkeiten
Digitale Strategie

Languages

Englisch (Professional Working)
Deutsch (Native or Bilingual)

Certifications

Certified Professional for Software Architecture Foundation Level
ITIL® Foundation
Develop GenAI Apps with Gemini and Streamlit Skill Badge
Servant Leadership – Dienende Führung
Experte für Computervalidierung

Publications

Parallel Processing via a Dual Olfactory Pathway in the Honeybee

Tobias Rosenbaum

IT meets Life Science

Mainz, Rheinland-Pfalz, Deutschland

Zusammenfassung

Driven by a passion for optimizing processes and solving complex challenges, I am a dynamic and results-oriented professional with a diverse background in biology, system analysis, software development, and system architecture. With a strong foundation in the regulated pharmaceutical environment, I excelled as a System Analyst at the Bavarian Blood Donation Service, where I facilitated software implementations while ensuring compliance with regulatory standards and holding patient security to the highest standards.

My expertise encompasses a wide range of functions, including requirement engineering, software development (C#), testing, rollout management, configuration management, and computer system validation. As the Quality Responsible Person, I consistently ensured the highest standards of quality and regulatory compliance within our department.

Transitioning to BioNTech, I took on the role of a Senior System Engineer, where I successfully built a team, implementing innovative ways of working. Recognizing my strategic acumen, I was promoted to the position of System Architect, where I provided valuable insights and guidance to drive the development and implementation of robust system architectures. In my subsequent role as a Team Lead of an Architecture Team, I focus on establishing effective communication structures, enabling both architects and application teams to deliver tailored software solutions for the company, particularly within the regulated GxP area.

I am dedicated to staying at the forefront of industry trends and technologies, continuously expanding my knowledge and skills to drive innovation and create value in every project I undertake.

Berufserfahrung

BioNTech SE

7 Jahre 2 Monate

Associate Director E2E Supply Chain Systems at BioNTech SE

Dezember 2024 - Present (1 Jahr)

As Associate Director E2E Supply Chain Management, I lead a team of highly motivated Engineers to deliver IT-systems designed for the unique challenges in providing individualized cancer treatments, e.g.: Needle to needle tracking.

My responsibilities include:

- Design of requirements in alignment with the Global Process Owner and the Domain System Architect
- Strategic direction setting, roadmap release planning and deployment of the IT solutions in scope
- Responsible for the application lifecycle management of its assigned system solutions, management of all changes and validation activities delivered by cross-functional team members and external implementation and service partners
- Responsible for the management of the governance process for the assigned function area and the alignment of priorities between business, technical organizations and interface partners
- Responsible for the management of the aligned team, talent development, recruiting of new team members, mission development and alignment within the organization and the stakeholders of the organization

Associate Director System Architecture Digital Solutions Quality

Mai 2021 - November 2024 (3 Jahre 7 Monate)

Empowering Innovation and Collaboration in System Architecture

Key Contributions:

- Empowered and supported architects in designing and delivering complex IT architectures in regulated environments.
- Established communication structures and frameworks to foster seamless coordination and knowledge sharing among architects.
- Facilitated the creation of architecture blueprints that align with BioNTech's goals and objectives, incorporating modern technologies and architectural patterns.
- Championed the use of LeanIX, JIRA, and Confluence to manage architecture documentation and streamline project management.

- Provided strategic guidance and mentorship to the architecture team, cultivating a culture of innovation and collaboration.
- Guided the development of a comprehensive GCLP area blueprint, ensuring alignment with regulatory requirements.
- Played a key role in preparing for upcoming audits, ensuring regulatory compliance and safeguarding patient safety.

Skills:

- Proficient in system architecture methodologies, including V-Model and UML
- Extensive experience in designing, implementing, and managing IT architectures in regulated environments
- Expertise in regulatory compliance and validation methodologies
- Proven ability to lead and mentor cross-functional teams
- Strong communication and collaboration skills

System Engineer

Oktober 2018 - Mai 2021 (2 Jahre 8 Monate)

Mainz und Umgebung, Deutschland

Elevating Laboratory Systems and Driving Innovation for Breakthrough Drugs

Key Contributions:

- Championed the redesign of the biobank system, introducing a modern architecture and streamlining data management processes.
- Spearheaded the technical aspects of our record breaking introduction of the global QC LIMS (7 months between project initiation and first go live in 3 sites), ensuring seamless integration with existing systems and enhancing data integrity.
- Established uniform processes across different LIM systems, promoting consistency and collaboration among teams.
- Collaborated effectively with the QC-LIMS vendor to develop an integrated validation approach, streamlining the validation process and facilitating timely deployment.
- Oversaw technical aspects of multiple projects, ensuring adherence to regulatory requirements and adherence to timelines.

Skills:

- Proficient in system engineering methodologies, including V-Model and UML

- Extensive experience in data management and integration across LIM systems
- Strong project management skills with a focus on quality, efficiency, and adherence to timelines
- Expertise in regulatory compliance and validation methodologies
- Proven ability to lead and manage cross-functional teams

Blutspendedienst des BRK

Systementwickler

August 2012 - September 2018 (6 Jahre 2 Monate)

Wiesentheid

Leading Software Implementation and Compliance Efforts in the Regulated Healthcare Industry

Key Contributions:

- Spearheaded the transition of our software development process from using legacy software (Delphi 5) to modern languages and environments (.NET with Visual Studio), improving efficiency, scalability, and maintainability of our systems.
- Led the implementation of the QC LIMS Maqsim Lab+, utilizing Delphi and .NET technologies for seamless integration with the existing central blood bank system Edgeblood.
- Served as the quality representative of the IT department, overseeing compliance and quality aspects in software development, and actively participating in supplier and regulatory body audits.
- Managed all phases of implementation projects, including requirements gathering, design, development, testing, rollout, and configuration management.
- Ensured comprehensive computer system validation (CSV) for all implemented systems, adhering to GxP guidelines and regulatory requirements.

Skills:

- Proficient in system analysis methodologies, including V-Model and UML
- Extensive experience in software development with Delphi and .NET (C# and VB.NET)
- In-depth understanding of regulated GxP environments and compliance requirements

- Proven project management skills with a focus on quality and adherence to timelines
- Expertise in data integrity and validation methodologies
- Strong leadership and communication skills

Roche

Systemanalytiker

Juni 2011 - Juli 2012 (1 Jahr 2 Monate)

As a System Analyst in the Automation Support Team at Roche Diagnostics, I played a pivotal role in optimizing batch release processes for the production of cancer treatments. I spearheaded the development of a centralized batch release cockpit that integrated data from disparate MES systems, providing real-time insights to support qualified personnel during the release process. This innovative solution streamlined workflows, reduced manual data entry, and enhanced regulatory compliance.

Key Contributions:

- Conceptualized, designed, and developed a validated dashboard for batch release automation
- Integrated data from multiple MES systems to provide a comprehensive overview of production processes
- Implemented data validation checks and automated workflow processes to ensure data integrity
- Collaborated with cross-functional teams to ensure alignment with regulatory requirements

Skills:

- Proficient in system analysis methodologies, including V-Model and UML
- Strong understanding of regulated GxP environments and compliance requirements
- Expertise in data visualization tools and dashboard development
- Excellent communication and collaboration skills

Julius-Maximilians Universität Würzburg

Research Assistant

August 2008 - Mai 2011 (2 Jahre 10 Monate)

Würzburg

During my academic journey, I had the opportunity to undertake a diploma thesis project that allowed me to combine my passion for biology and technology. For my thesis, I developed a sophisticated system to record and analyze neuronal activity from honey bees. Utilizing my expertise in programming and machine learning, I also created a set of algorithms in MATLAB to recognize patterns in the neuronal recordings.

Dade Behring Marburg GmbH
Software Test Engineer
März 2005 - September 2005 (7 Monate)
Schwalbach

After completing my apprenticeship as a computer system developer, I began my professional career by working as a software tester for a Laboratory Information Management System (LIMS) that specialized in working with blood analyzers. This role allowed me to gain practical experience in ensuring the quality and reliability of software solutions within the regulated healthcare environment.

Ausbildung

Julius-Maximilians-Universität Würzburg
Diplom, Biologie, allgemein · (2005 - 2011)

Provadis - Partner für Bildung und Beratung GmbH
Fachinformatiker, Anwendungsentwicklung · (2002 - 2005)